a)

$$G_H(s) = \frac{1 - e^{-sT}}{s}$$

$$G_{AH}(j\omega) = \frac{1 - e^{-j\omega T}}{j\omega \cdot T} = \frac{e^{\frac{T}{2}j\omega} - e^{-\frac{T}{2}j\omega}}{2 \cdot \frac{1}{2} j\omega} \cdot e^{-\frac{T}{2}j\omega}$$

mit $\sin\left(\frac{\omega \cdot T}{2}\right) = \frac{e^{\frac{T}{2}j\omega} - e^{-\frac{T}{2}j\omega}}{2j}$

$$G_{AH}(j\omega) = \frac{\sin\left(\frac{\omega \cdot T}{2}\right)}{\frac{1}{2}\omega \cdot T} \cdot e^{-\frac{T}{2}j\omega} \approx e^{-\frac{T}{2}j\omega}$$

$$G_{AH}(s) = e^{-\frac{T}{2}s} \approx \frac{1}{1 + \frac{T}{2}s}$$

$$G_{AH}(s) \cdot G_{Si}(s) = \frac{1}{1 + \frac{T}{2}s} \cdot \frac{1}{1 + T_{EM} \cdot s} = \frac{1}{1 + \left(\frac{T}{2} + T_{EM}\right) \cdot s + \frac{T}{2} \cdot T_{EM} s^2}$$

$\frac{T}{2} \cdot T_{EM} \cdot s^2$ ist vernachlässigbar

$$G_{AH}(s) \cdot G_{Si}(s) \approx \frac{1}{1 + \left(\frac{T}{2} + T_{EM}\right) s} = \frac{1}{1 + 3{,}5 \times 10^{-3}\, s}$$

(b)

$$G_0(s) = G_{AH}(s) \cdot G_R(s) \cdot G_M(s) \cdot G_{Si}(s)$$

$$= \frac{1}{1 + \frac{T}{2}s} \cdot \frac{K_P(1 + T_n s)}{T_n \cdot s} \cdot \frac{1}{1 + T_{EM} \cdot s} \cdot \frac{1}{J s}$$

$$G_W(s) = \frac{G_0(s)}{1 + G_0(s)} = \frac{K_P(1 + T_n s)}{\left[1 + \left(\frac{T}{2} + T_{EM}\right) s\right] \cdot T_n J s^2 + K_P(1 + T_n s)}$$

$$G_{FW}(s) = G_W(s) \cdot G_F(s) = G_W(s) \cdot \frac{1}{1 + T_F s}$$

mit $T_N = T_F$.

$$G_{FW}(s) = \frac{K_P}{T_n J s^2 + \left(\frac{T}{2} + T_{EM}\right) T_n J s^3 + K_P(1 + T_n s)}$$

$$= \frac{1}{1 + T_n s + \frac{T_n}{K_P} J s^2 + \frac{\left(\frac{T}{2} + T_{EM}\right) T_n}{K_P} J s^3}$$

$$|G_{FW}(j\omega)|^2 = G_{FW}(j\omega) \cdot G_{FW}(-j\omega)$$

$$G_{FW}(j\omega) = \frac{1}{1 + j T_n \omega - \frac{T_n}{K_P} J \omega^2 - j \frac{(\frac{T}{2} + T_{EM}) \cdot T_n}{K_P} J \omega^3}$$

$$= \frac{1}{\left(1 - \frac{T_n}{K_P} J \omega^2\right) + j \left(T_n \omega - \frac{(\frac{T}{2} + T_{EM}) \cdot T_n}{K_P} J \omega^3\right)}$$

$$G_{FW}(-j\omega) = \frac{1}{\left(1 - \frac{T_n}{K_P} J \omega^2\right) - j \left(T_n \omega - \frac{(\frac{T}{2} + T_{EM}) \cdot T_n}{K_P} J \omega^3\right)}$$

$$|G_{FW}(j\omega)|^2 = \frac{1}{\left(1 - \frac{T_n}{K_P} J \omega^2\right)^2 + \left(T_n \omega - \frac{(\frac{T}{2} + T_{EM}) \cdot T_n}{K_P} J \omega^3\right)^2}$$

$$= \frac{1}{1 - 2\frac{T_n}{K_P} J \omega^2 + \frac{T_n^2}{K_P^2} J^2 \omega^4 + T_n^2 \omega^2 - 2 \frac{T_n^2 (\frac{T}{2} + T_{EM})}{K_P} J \omega^4 + \frac{(\frac{T}{2} + T_{EM})^2 T_n^2}{K_P^2} J^2 \omega^6}$$

$$= \frac{1}{1 + \left(T_n^2 - 2\frac{T_n}{K_P} J\right)\omega^2 + \left(\frac{T_n^2}{K_P^2} J^2 - 2 \frac{T_n^2 (\frac{T}{2} + T_{EM})}{K_P} J\right)\omega^4 + \frac{(\frac{T}{2} + T_{EM})^2 T_n^2}{K_P^2} J^2 \omega^6}$$

Für Betragsoptimums ist $|G(j\omega)|^2 = 1$

$$\begin{cases} T_n^2 - 2 \frac{T_n}{K_P} J = 0 \\ \frac{T_n^2}{K_P^2} J^2 - 2 \frac{T_n^2 (\frac{T}{2} + T_{EM})}{K_P} J = 0 \end{cases} \Rightarrow \begin{cases} T_n = 0{,}014 \\ K_P = 0{,}06 \end{cases}$$

c)

Rechteckregel.

1) Pol- Nullstellen- Form

$$G_R(s) = K_P \left(1 + \frac{1}{T_n s}\right)$$

$$\longrightarrow u(t) = K_P \cdot e(t) + \frac{K_P}{T_n} \int_0^t e(\tau)\, d\tau$$

$t = kT$

$$u_k = K_P \cdot e_k + \frac{K_P}{T_n} \int_0^{(k-1)T} e(\tau)\, d\tau + \frac{K_P}{T_n} \int_{(k-1)T}^{kT} e(\tau)\, d\tau$$

$$u_{k-1} = K_P \cdot e_{k-1} + \underbrace{\frac{K_P}{T_n} \int_0^{(k-1)T} e(\tau)\, d\tau}_{u_{I,k-1}}$$

$$u_{I,k-1} = u_{k-1} - K_P \cdot e_{k-1}$$

$$u_k = K_P \cdot e_k - K_P e_{k-1} + u_{k-1} + \frac{K_P}{T_n} \underbrace{\int_{(k-1)T}^{kT} e(\tau)\, d\tau}_{e_k \cdot T}$$

$$u_k = K_P \left(1 + \frac{T}{T_n}\right) e_k - K_P \cdot e_{k-1} + u_{k-1}$$

$$u_k = 0.06428 \cdot e_k - 0{,}06 \cdot e_{k-1} + u_{k-1}$$

2) Summen- Form

$$G_R(s) = K_P \left(1 + \frac{1}{T_n s}\right)$$

$$u(t) = K_P \cdot e(t) + \frac{K_P}{T_n} \int_0^t e(\tau)\, d\tau$$

$t = kT$

$$u_k = K_P \cdot e_k + \int_0^{kT} e(\tau)\, d\tau = K_P \cdot e_k + u_{I,k}$$

$$u_{I,k} = u_{I,k-1} + \frac{K_P}{T_n} \int_{(k-1)T}^{kT} e(\tau)\, d\tau = u_{I,k-1} + \frac{K_P}{T_n} e_k \cdot T$$

$$\begin{cases} u_k = K_P \cdot e_k + u_{I,k} \\ u_{I,k} = u_{I,k-1} + \frac{K_P}{T_n} e_k \cdot T \end{cases} \Longrightarrow \begin{cases} u_k = 0.06\, e_k + u_{I,k} \\ u_{I,k} = u_{I,k-1} + 0.0043 \cdot e_k \end{cases}$$

d) Trapezregel

1) Pol-Nullstellen-Form

$$G_R(s) = k_P\left(1 + \frac{1}{T_n s}\right)$$

$$u(t) = k_P \cdot e(t) + \frac{k_P}{T_n}\int_0^t e(\tau)\,d\tau.$$

mit $t = kT$

$$u_k = k_P \cdot e_k + \frac{k_P}{T_n}\int_0^{(k-1)T} e(\tau)\,d\tau + \frac{k_P}{T_n}\int_{(k-1)T}^{kT} e(\tau)\,d\tau.$$

$$u_{k-1} = k_P \cdot e_{k-1} + \underbrace{\frac{k_P}{T_n}\int_0^{(k-1)T} e(\tau)\,d\tau}_{u_{I,k-1}}$$

$$u_{I,k-1} = u_{k-1} - k_P \cdot e_{k-1}$$

$$u_k = k_P e_k + u_{k-1} - k_P e_{k-1} + \frac{k_P}{T_n}\underbrace{\int_{(k-1)T}^{kT} e(\tau)\,d\tau}_{\frac{T}{2}(e_{k-1}+e_k)}$$

$$= \left(k_P + \frac{T}{2}\cdot\frac{k_P}{T_n}\right)e_k + \left(\frac{k_P}{T_n}\cdot\frac{T}{2} - k_P\right)e_{k-1} + u_{k-1}$$

$$= 0{,}06214\, e_k + 0{,}05786\, e_{k-1} + u_{k-1}$$

2) Summen-Form

$$G_R(s) = k_P\left(1 + \frac{1}{T_n s}\right)$$

$$u(t) = k_P \cdot e(t) + \frac{k_P}{T_n}\int_0^t e(\tau)\,d\tau.$$

$t = kT$

$$u_k = k_P \cdot e_k + \frac{k_P}{T_n}\int_0^{kT} e(\tau)\,d\tau$$

$$u_{I,k} = \frac{k_P}{T_n}\int_0^{(k-1)T} e(\tau)\,d\tau + \frac{k_P}{T_n}\int_{(k-1)T}^{kT} e(\tau)\,d\tau$$

$$= u_{I,k-1} + \frac{k_P}{T_n}\cdot\frac{T}{2}(e_{k-1} + e_k)$$

$$\begin{cases} u_k = k_P e_k + u_{I,k} \\ u_{I,k} = u_{I,k-1} + \frac{T}{2}\cdot\frac{k_P}{T_n}(e_{k-1} + e_k) \end{cases} \Longrightarrow \begin{cases} u_k = 0{,}06\, e_k + u_{I,k} \\ u_{I,k} = u_{I,k-1} + 0{,}0214(e_{k-1} + e_k) \end{cases}$$